

An exhaustive $\Delta\Delta G$ scan of human α -galactosidase A from structure-model embeddings

Experiment 10 · ddG_with_Boltz · Boltz-2 pair-track embeddings · label-free application

The $\Delta\Delta G$ predictor built on Boltz-2 embeddings is applied the way it would be used in practice: pointed at a single protein of interest with **no experimental labels**, and asked to score every possible point mutation. On human α -galactosidase A we score **2,238 substitutions across 177 positions** and compare against an independent FoldX scan of the same protein. The two agree at **Spearman $\rho = +0.595$** , but the agreement is strongly residue-dependent — **+0.639 at non-glycine sites versus +0.458 at glycines** — and the two methods differ by 6.5 $\times$ in dynamic range. The scan behaves as a **destabilization ranker**: only 7.7% of its predictions fall below zero.

1. Motivation

Stability predictors are normally reported on curated benchmarks, where every variant has a measured $\Delta\Delta G$. The practical question is different: given one protein and no measurements, can the model rank its mutational landscape usefully? Answering it requires an exhaustive scan — every position, every substitution — and a way to judge the output when no ground truth exists.

Human α -galactosidase A is a demanding test case. It is a 398-residue human lysosomal enzyme, far from the small designed domains that dominate the training corpus, and it is absent from every dataset the model was trained on, so its predictions here are genuinely blind. An independent FoldX scan of the same protein provides a second, mechanistically unrelated opinion.

2. Methods

2.1 Representation and model

For each mutation the wild-type and mutant sequences are passed through Boltz-2 in an embeddings-only mode, and the pair-track (z) row of the mutated residue is pooled over partners for each. The two pooled vectors are **concatenated** (128 + 128), retaining both absolute levels rather than only their difference. The regressor is a 5-seed ensemble of MLPs (256–128–64) over median-imputed, standardised features, trained with **antisymmetry augmentation**: the reverse mutation is the two halves exchanged with the label negated.

2.2 Three training regimes

Because no labels exist for the target, the model must be trained elsewhere. Three regimes are reported together, differing only in training distribution: **A** — a mega-scale folding-assay corpus of mostly small domains; **B** — a literature-curated corpus of natural proteins; **D** — A pretrained then fine-tuned on B. They are a robustness check on that free parameter rather than three competing answers: the training corpus is the one modelling choice that no measurement on this target can adjudicate, so every headline result is repeated under each regime in §3.2 (Table 2) and the value reported everywhere else is their mean. Positive $\Delta\Delta G$ denotes destabilization throughout.

2.3 Comparison metric

Predicted and FoldX values are compared by **rank correlation**. This is not a stylistic choice: FoldX holds the backbone rigid, so a substitution that would be accommodated by a small conformational shift is instead scored as an unrelievable clash, producing values that reach tens of kcal/mol. Those points dominate any squared-error or Pearson statistic. Pearson is reported alongside, on raw and clipped values, for reference only.

3. Results

3.1 The predicted landscape

The three regimes place the landscape at similar levels (mean $\Delta\Delta G$ +1.28, +1.17 and +1.33 kcal/mol for A, B and D) and agree closely on ordering (pairwise Pearson 0.78–0.92), with a mean across-regime spread of 0.29 kcal/mol. Conclusions below are therefore not artifacts of one training corpus. **Every predicted value reported from here on is the mean across the three regimes** — the quantity plotted, tabulated and correlated in §3.2–§3.5. Table 2 repeats the headline comparison under each regime separately, so the corpus dependence can be checked rather than assumed.

Within that landscape the dominant axis is **position, not substituted residue**: position accounts for $\omega^2 = 0.60$ of the variance in predicted $\Delta\Delta G$ against $\omega^2 = 0.12$ for the identity of the incoming residue. (The bias-corrected ω^2 is used because the two factors differ nine-fold in group count — 177 positions against 20 residues — which inflates the uncorrected figure for position.) So the model is reading the site far more than the substitution.

It grades those sites by degree rather than sorting them into tolerant and intolerant classes. Across the 123 positions where at least 10 of the 19 substitutions were scored, the per-position mean $\Delta\Delta G$ spans -0.10 to +3.31 kcal/mol — a wide range, but a near-continuous one. Positions that are absolute in either direction are rare: only 2 destabilise on *every* scored substitution (all above +1.5 kcal/mol) and 1 tolerates all of them (all below +0.5). Restricting to still better-covered positions removes them entirely, so apparent all-or-nothing sites in this scan are an artifact of thin coverage rather than a feature of the predictions.

3.2 Agreement with an independent predictor

Subset	n	Spearman ρ	Pearson (raw)	Pearson (clipped)
All scored substitutions	2,238	+0.595	+0.350	+0.521
Non-glycine sites	1,733	+0.639	—	—
Glycine sites	505	+0.458	—	—
Excluding FoldX clash regime (<10 kcal/mol)	2,043	+0.602	—	—

Table 1. Rank agreement between the embedding scan and FoldX, the predicted value being the across-regime mean of \$3.1. Pearson values are shown for completeness; the clipped column bounds FoldX at ± 10 kcal/mol.

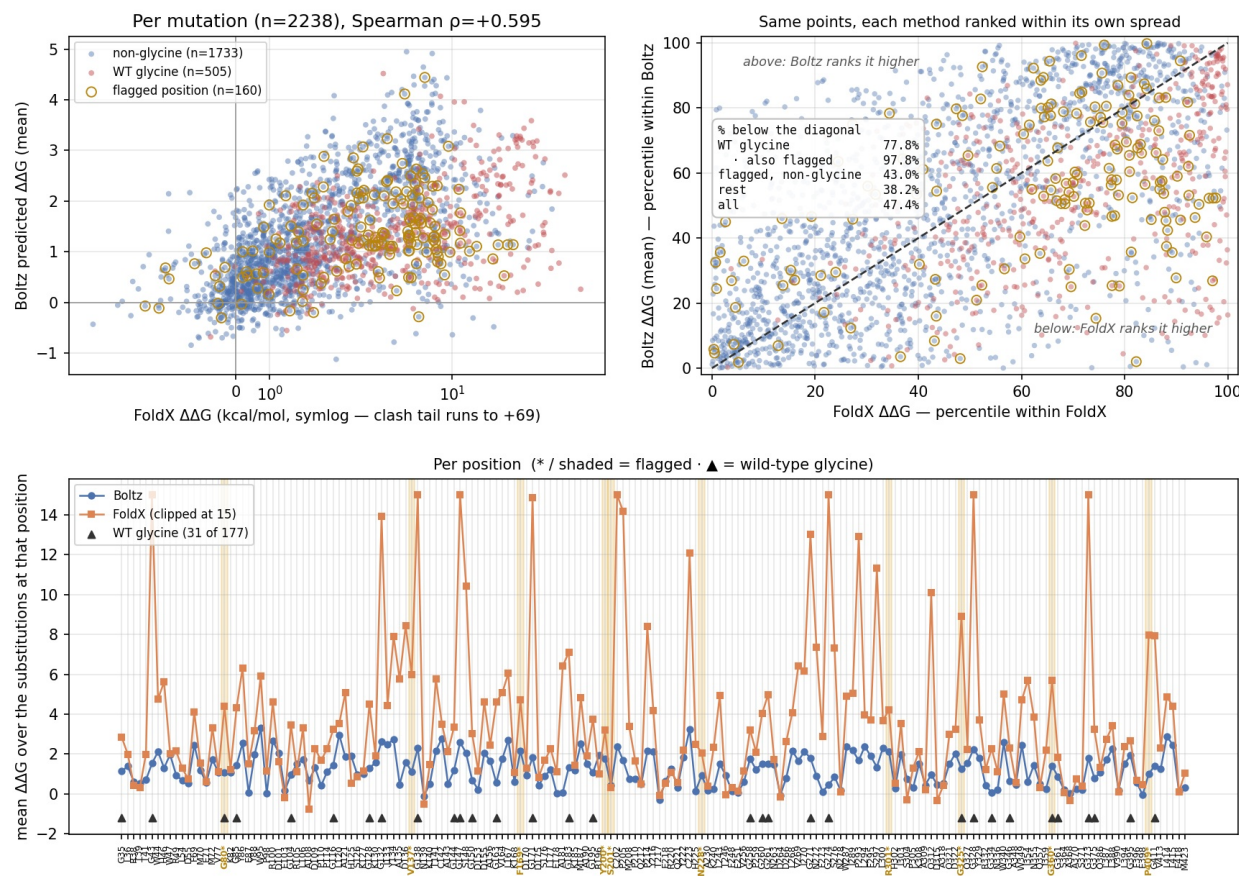


Figure 1. Predicted $\Delta\Delta G$ (across-regime mean) against FoldX. Top left: per-substitution comparison; the horizontal axis is symlog because FoldX extends to tens of kcal/mol. Top right: the same points with each method ranked *within its own spread*, so the diagonal needs no fitting and is immune to the dynamic-range gap; a point below it is one FoldX ranks higher than the embedding model does. Glycines fall below it 78% of the time against 38% for the rest, and flagged positions that *are not* glycine sit at 43%, i.e. within noise of the rest. Bottom: per-position means. Agreement is visibly tighter away from glycine positions.

The headline agreement of $\rho = +0.595$ is essentially unchanged when FoldX's clash regime is excluded (+0.602), so it does not rest on the two methods merely concurring that clashes are bad. It is, however, **strongly residue-dependent**: at non-glycine sites the two predictors agree at +0.639, at glycines only +0.458. Glycine is the residue whose replacement most often demands backbone accommodation, and it is where the two methods' treatments diverge most.

Training regime	ρ all	ρ non-Gly	ρ Gly	stabilizing
A — Tsuboyama-only	+0.620	+0.661	+0.461	6.2%
B — FireProt-only	+0.507	+0.561	+0.402	11.8%
D — A fine-tuned on B	+0.593	+0.638	+0.438	9.4%
Mean of A, B, D (used throughout)	+0.595	+0.639	+0.458	7.7%

Table 2. The same comparison computed separately under each training regime. The mean row is the value used everywhere else in this report.

Neither of the report's two central claims depends on which corpus the model was trained on. Agreement with FoldX varies between $\rho = +0.507$ and +0.620 across the three regimes, spanning the reported +0.595; the glycine deficit appears in *every* regime, with non-glycine agreement exceeding glycine agreement in each row; and the stabilizing tail stays small throughout (6.2–11.8% of predictions below zero). Regime B agrees with FoldX least, but that ordering should not be read as an accuracy ranking: on labelled external benchmarks the same predictor ranks these regimes the other way round, and agreement with FoldX on a glycine-rich set is not a measure of correctness in the absence of ground truth.

3.3 The methods disagree about magnitude far more than about order

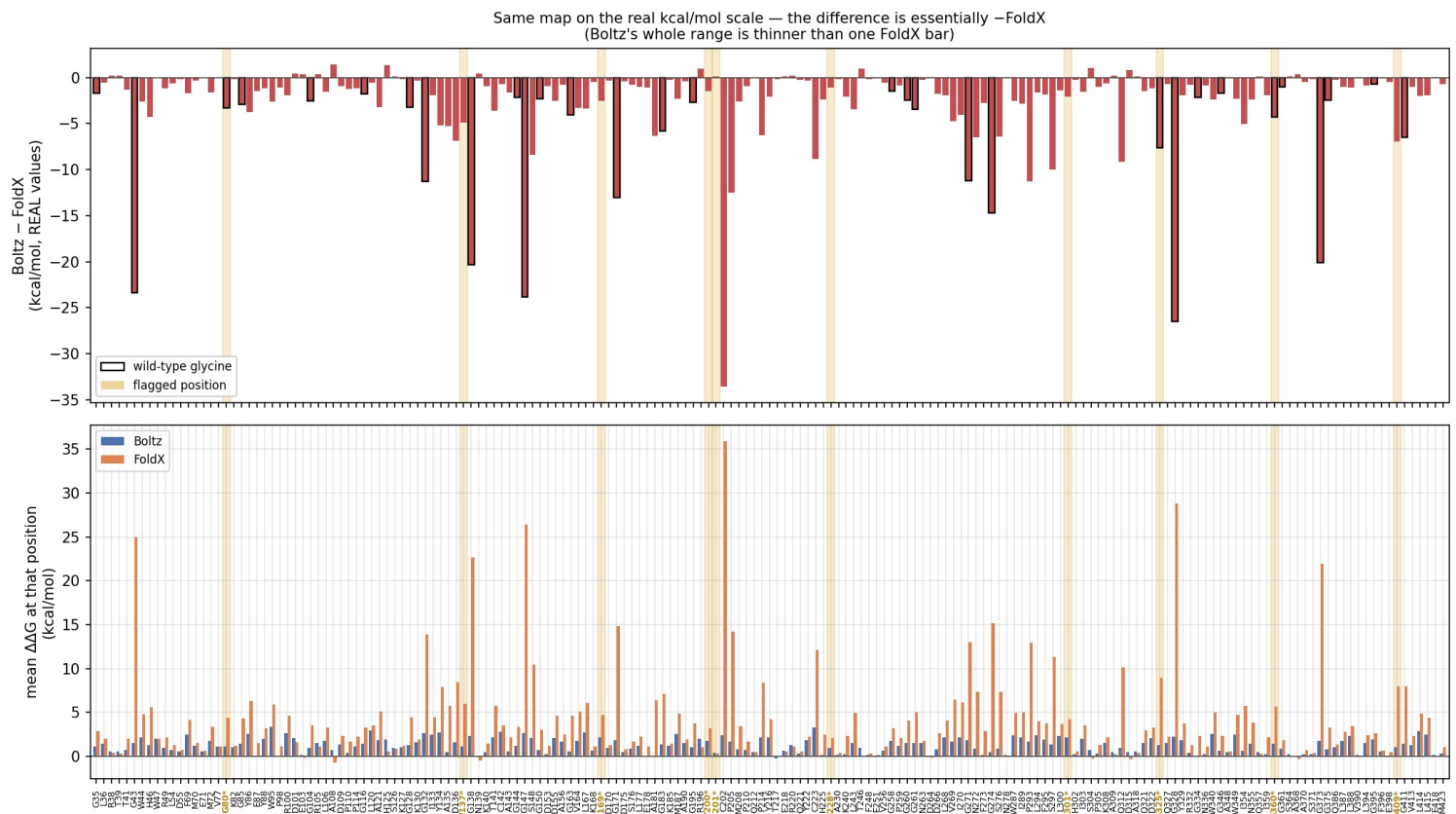


Figure 2. The same comparison in real units. Top: per-position difference. Bottom: per-position means of both methods on one axis — the narrow bars are the embedding model's entire dynamic range.

Per-position variability is 5.36 kcal/mol for FoldX against 0.82 for the embedding model, a factor of 6.5. The raw difference between the two correlates with FoldX alone at $p = -0.948$, i.e. it is very nearly the negative of the FoldX value and carries almost no independent information. Magnitude, not ordering, is where the two methods part company — and rank statistics are the only ones that survive it.

3.4 A hypothesis from structural inspection

Ten positions had been flagged, on structural grounds, as sites where the model appeared to overestimate destabilization. Those positions do agree with FoldX less well than the remainder (median per-position $p + 0.261$ versus $+0.404$), and glycine positions likewise ($+0.285$ versus $+0.420$). Neither separation reaches significance at the position level ($p = 0.120$ and $p = 0.058$, Mann-Whitney), because per-position correlations over ~ 19 substitutions are noisy. The glycine effect is unambiguous only at the substitution level, where the sample is large.

At that level the two hypotheses turn out not to be independent. Ranking each method within its own spread (Figure 1, top right), glycine substitutions fall on the FoldX-higher side of the diagonal 77.8% of the time versus 38.2% for the rest — but flagged positions that are not glycines sit at 43.0%, indistinguishable from that baseline, while the flagged positions that *are* glycines reach 97.8%. The flagged-position hypothesis is therefore largely a restatement of the glycine effect, with a genuine additional shift only where the two coincide.

3.5 An external check against measured data

No measured $\Delta\Delta G$ exists for this protein, but a measured *proxy* does. A functional study of Fabry disease expressed 157 missense variants of this enzyme in human cells and quantified the residual α -galactosidase activity of each (Lukas et al., *PLoS Genetics* 2013). Activity is not stability — a substitution at a catalytic residue abolishes turnover in a perfectly folded protein — so **variants at active-site positions are excluded** here, leaving 41 that the scan also covers. What remains is a weak but genuine external constraint: a variant destabilised enough to misfold should not retain activity.

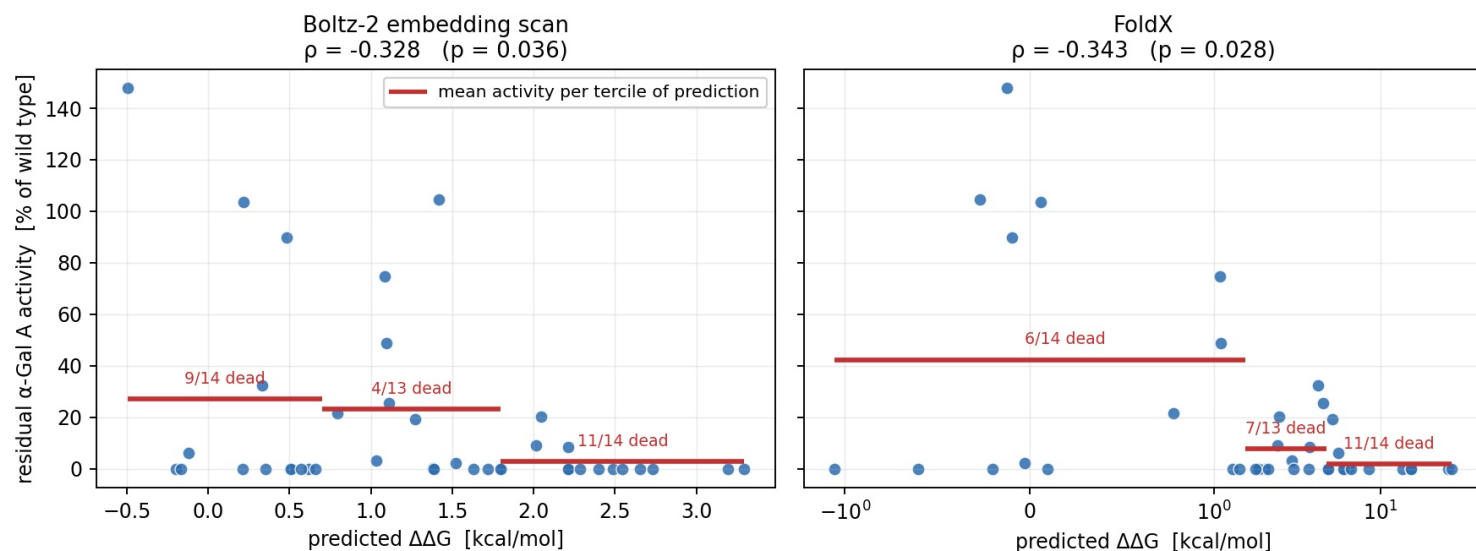


Figure 3. Predicted $\Delta\Delta G$ against measured residual activity for the 41 variants outside the active site. Red bars are the mean activity within tertiles of the prediction, labelled with how many variants in each tertile have exactly zero activity; the FoldX axis is symlog. Both methods rank in the correct direction and neither separates the many dead variants from each other — activity saturates at zero long before $\Delta\Delta G$ does.

The predicted ranking runs the right way: $\rho = -0.328$ ($p = 0.036$) against measured activity, versus -0.343 ($p = 0.028$) for FoldX on the same variants — a difference indistinguishable from zero (paired bootstrap CI [-0.250, +0.286]). Variants with no detectable activity (24 of 41) carry a higher predicted $\Delta\Delta G$ than those retaining any (+1.76 versus +1.10 kcal/mol median, Mann-Whitney $p = 0.042$; the same test on FoldX gives $p = 0.061$).

The size of the effect is the honest part of this result. A correlation near 0.33 against a saturating, zero-inflated readout on 41 variants establishes that the scan ranks loss of function better than chance; it does not establish accuracy on a $\Delta\Delta G$ scale, and it cannot, because the measurement is not one.

4. Interpretation

The glycine deficit is a property of the model, not of this protein. An independent error analysis of the same predictor on labelled held-out data finds buried glycines to be one of its few genuine weak spots once effect size is controlled for, alongside proline targets. The present scan reproduces that signature blind, on a protein with no measurements at all, which is the stronger form of the observation.

Burial degrades the physics-based method, not the learned one. The extreme FoldX values here occur at buried glycines, where a rigid backbone cannot relieve the clash. The same error analysis finds the embedding model's *relative* accuracy to be flat across burial tertiles. The two methods therefore fail in different places, which is what makes their disagreement informative even without ground truth.

The practical limitation is the stabilizing tail. Only 173 of 2,238 predictions (7.7%) fall below zero and just 20 below -0.5 kcal/mol. The model is a destabilization ranker: it is well suited to asking which substitutions a protein will not tolerate, and poorly suited to searching for stabilizing ones — the regime that matters most for engineering.

What this comparison cannot establish. No measured $\Delta\Delta G$ exists for this protein — the activity data of §3.5 constrains loss of function, not folding free energy — so FoldX is a second opinion and not a reference. Where the two disagree, neither is thereby shown to be wrong. The one asymmetry available is mechanistic: in the clash regime there is an independent reason to distrust FoldX, whereas at exposed sites the disagreement is simply unresolved.

5. Limitations

- **Partial coverage.** 2,238 substitutions at 177 of the protein's 398 positions. The scored positions are not a random sample, so protein-wide summaries should not be read off them.
- **No ground truth.** Neither absolute accuracy nor calibration can be assessed on this target; only agreement with another predictor, and a rank check against a measured *activity* proxy that saturates at zero and confounds folding with catalysis.
- **Per-position statistics are noisy.** With ~ 19 substitutions per site, individual per-position correlations carry wide uncertainty and should not be ranked against each other.
- **Rank-based comparison is deliberately blind to scale.** It cannot see the $6.5\times$ dynamic-range difference, which is reported separately.

6. Conclusion

A structure-model embedding predictor can be applied to a whole protein without labels and produces a mutational landscape that agrees with an established physics-based method at $\rho = +0.595$, rising to $+0.639$ away from glycine sites. The two methods fail in different, identifiable places: FoldX at buried glycines, where its rigid backbone forces runaway clash energies, and the embedding model on stabilizing mutations, which it rarely predicts at all. For triaging which substitutions a protein will not tolerate, the scan is usable today; for discovering stabilizing ones, its blind spot is the limiting factor.